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Electric power control system and power supply system

Abstract

An electric power control system includes: a bidirectional electric power converter including AC and DC terminals, the AC terminal being connected to an AC power system, the DC terminal being connected to a DC power system to which one or more DC power loads are connected, the bidirectional electric power converter mutually converting AC power of the AC power system and DC power of the DC power system; an electric power storage that is connected to the DC power system and stores electric power of the DC power system; and a bidirectional electric power conversion controller that performs first control of controlling an output or input of the AC terminal of the bidirectional electric power converter such that a charge current, discharge current, charge electric power, or discharge electric power of the electric power storage coincides with a predetermined target current value or a predetermined target electric power value.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS

(1) This is a National Phase of International Patent Application No. PCT/JP2020/026868 filed Jul. 9, 2020, which in turn claims the benefit of Japanese Patent Application No. 2019-152471 filed Aug. 23, 2019. The disclosure of the prior applications is hereby incorporated by reference herein in its entirety.

TECHNICAL FIELD

(2) The present invention relates to an electric power control system and a power supply system.

BACKGROUND ART

(3) Known as a conventional electric power control system is a method of controlling a propulsion system of a movable body, the method being able to, when short-cycle load fluctuation occurs, suppress an influence of the load fluctuation on an electric power system (see PTL 1).

(4) The method of controlling the propulsion system of the movable body includes: subjecting a first power converter to drooping control based on a first electric power command value such that a point indicated by a frequency of the electric power system and system supplied/received electric power is located on a drooping characteristic line showing a relation between a target value of the frequency of the electric power system and a target value of the system supplied/received electric power; and controlling a second power converter based on a second electric power command value such that electric power converted by the second power converter becomes the second command value. The first electric power command value is generated by removing a fluctuation component from an original electric power command value that is a commanded electric power value or an electric power command value generated based on a commanded revolution value of a motor generator. The second electric power command value is based on the original electric power command value.

CITATION LIST

Patent Literature

(5) PTL 1: Japanese Laid-Open Patent Application Publication No. 2016-055850 (see the abstract)

SUMMARY OF INVENTION

Technical Problem

(6) In the above conventional method of controlling the propulsion system of the movable body, a DC power load connected to a DC terminal of the first power converter needs to be a special load that is a motor generator connected to the DC terminal of the first power converter through the

second power converter. Therefore, PTL 1 does not describe a method of controlling a propulsion system of a movable body, the method being able to suppress an influence of load fluctuation on an (AC) electric power system when: a general DC power load other than a motor generator connected through a (second) electric power converter (hereinafter simply referred to as a “general DC power load”) is connected to a DC terminal of a (first) electric power converter including an AC terminal connected to an AC power system; and short-cycle (sudden) load fluctuation occurs at the general DC power load.

(7) The present invention was made to solve the above problem, and an object of the present invention is to provide an electric power control system and a power supply system, each of which includes a bidirectional electric power converter connected between an AC power system and a DC power system and is able to, when sudden load fluctuation occurs at a general DC power load, suppress an influence of the load fluctuation on the AC power system.

Solution to Problem

(8) An electric power control system according to an aspect of the present invention includes: a bidirectional electric power converter including an AC terminal and a DC terminal, the AC terminal being connected to an AC power system, the DC terminal being connected to a DC power system to which one or more DC power loads are connected, the bidirectional electric power converter mutually converting AC power of the AC power system and DC power of the DC power system; an electric power storage that is connected to the DC power system and stores electric power of the DC power system; and a bidirectional electric power conversion controller that performs first control of controlling an output or input of the AC terminal of the bidirectional electric power converter such that a charge current, discharge current, charge electric power, or discharge electric power of the electric power storage coincides with a predetermined target current value or a predetermined target electric power value.

Advantageous Effects of Invention

(9) The present invention can provide an electric power control system and a power supply system, each of which includes a bidirectional electric power converter connected between an AC power system and a DC power system and is able to, when sudden load fluctuation occurs at a general DC power load, suppress an influence of the load fluctuation on the AC power system.

Description

BRIEF DESCRIPTION OF DRAWINGS

(1) FIG. 1 is a block diagram showing the configuration of a power supply system according to Embodiment 1 of the present application.

(2) FIG. 2 is a block diagram showing the configuration related to electric power conversion control of an inverter controller of FIG. 1.

(3) FIG. 3 is a block diagram showing the configuration of a converter controller of FIG. 1.

(4) FIG. 4 is a graph showing the result of a simulation of the power supply system of FIG. 1.

(5) FIG. 5 is a graph showing the result of a simulation of the power supply system of Comparative Example.

(6) FIG. 6 is a block diagram showing the configuration of the power supply system according to Embodiment 2 of the present application.

(7) FIG. 7 is a block diagram showing the configuration of the inverter controller of FIG. 6.

(8) FIG. 8 is a graph showing the result of a simulation of the power supply system of FIG. 6.

(9) FIG. 9 is a graph showing the result of a simulation of the power supply system of Comparative Example.

(10) FIG. 10 is a flow chart showing a dead zone operation of a battery current controller in the inverter controller of the power supply system according to Embodiment 3 of the present

application.

(11) FIG. **11** is a graph showing the result of a simulation of the power supply system of Embodiment 3.

(12) FIG. **12** is a block diagram showing the configuration of the inverter controller of the power supply system according to Embodiment 4 of the present application.

(13) FIG. **13** is a block diagram showing the configuration of the inverter controller of the power supply system according to Embodiment 5 of the present application.

(14) FIG. **14** is a block diagram showing the configuration of the inverter controller of the power supply system according to Embodiment 6 of the present application.

DESCRIPTION OF EMBODIMENTS

(15) Hereinafter, embodiments of the present application will be described with reference to the drawings. In the following description and the drawings, the same reference signs are used for the same or corresponding components, and the repetition of the same explanation is avoided. The present application is not limited to the embodiments below.

Embodiment 1

Configuration

(16) FIG. **1** is a block diagram showing the configuration of a power supply system according to Embodiment 1 of the present application. As shown in FIG. **1**, a power supply system **200** of Embodiment 1 includes an AC power system **10**, a DC power system **20**, and an electric power control system **100**.

(17) The power supply system **200** may be a microgrid including an AC power system and a DC power system. The AC power system may be a commercial system (an electric power system of an electric power company). Examples of the power supply system **200** include: electric power supply systems of ships, vehicles, aircrafts, and helicopters; and microgrids. An installation location of the microgrid is not limited to a position on a movable body and may be a position on ground (land).

(18) Herein, the AC power system **10** transmits three-phase AC power. The AC power system **10** includes a bus line **13**. For example, a power generator **12** and one or more general AC power loads (hereinafter abbreviated as “AC loads”) **14** are connected to the bus line **13**. A driving source of the power generator **12** is not especially limited. Herein, the driving source of the power generator **12** is an engine **11**. FIG. **1** shows only one AC load **14**. Examples of the AC loads **14** include various electrical apparatuses and lighting apparatuses.

(19) The DC power system **20** includes a bus line **21**. For example, one or more general DC power loads (hereinafter abbreviated as “DC loads”) **22** are connected to the bus line **21**, and an AC motor **24** is connected to the bus line **21** through a load inverter **23**. FIG. **1** shows only one DC load **22**. Examples of the DC loads include: electrical apparatuses for both AC and DC; lighting apparatuses; and DC motors. Hereinafter, the DC load **22**, the load inverter **23**, and the AC motor **24** will be collectively called a DC power load (**22** to **24**) and will be described together.

(20) The electric power control system **100** includes: a bidirectional electric power converter (**31**) including an AC terminal and a DC terminal, the AC terminal being connected to the AC power system, the DC terminal being connected to the DC power system **20** to which one or more DC power loads (**22** to **24**) are connected, the bidirectional electric power converter (**31**) mutually converting AC power of the AC power system **10** and DC power of the DC power system **20**; an electric power storage (**25**, **26**) that is connected to the DC power system **20** and stores electric power of the DC power system **20**; and a bidirectional electric power conversion controller (**32**) that performs first control of controlling an output or input of the AC terminal of the bidirectional electric power converter (**31**) such that a charge current, discharge current, charge electric power, or discharge electric power of the electric power storage (**25**, **26**) coincides with a predetermined target current value I_{Bt} (see FIG. **2**) or a predetermined target electric power value. As an example, the following will describe a case where as the first control, the bidirectional electric power conversion controller (**32**) controls the output or input of the AC terminal of the bidirectional

electric power converter (31) such that the charge current or discharge current of the electric power storage (25, 26) coincides with the predetermined target current value IBt (see FIG. 2).

(21) A case where as the first control, the bidirectional electric power conversion controller (32) controls the output or input of the AC terminal of the bidirectional electric power converter (31) such that the charge electric power or discharge electric power of the electric power storage (25, 26) coincides with the predetermined target electric power value can also be performed in the same manner as the following case with reference to the following description.

(22) Specifically, the electric power control system 100 includes a converter 25, a battery 26, a power supply inverter 31, an inverter controller 32, a converter controller 33, a battery ammeter 34, a battery voltmeter 35, and a DC power system voltmeter 36. The converter 25 and the battery 26 constitute an electric power storage that stores DC power of the DC power system 20.

(23) The battery 26 is one example of an electric power storage device. Examples of the electric power storage device includes batteries (secondary batteries) and capacitors. Herein, the battery is used as the electric power storage device. The battery 26 is connected to the bus line 21 of the DC power system 20 through the converter 25. The battery 26 is disposed to suppress the fluctuation of the voltage of the DC power system 20. Specifications (capacity, rated voltage, etc.) of the battery 26 are suitably determined in accordance with a reference voltage, capacity, and the like of the DC power system 20.

(24) The converter 25 is constituted by, for example, a DC/DC converter. The converter 25 is controlled by the converter controller 33 to convert the voltage of the battery 26 to a predetermined voltage (target voltage value) by a voltage increase or a voltage decrease. Herein, the target voltage value is the reference voltage of the DC power system 20. The converter (DC/DC converter) 25 is constituted by, for example, a DC chopper, and changes a duty ratio to control a ratio of the voltage increase or the voltage decrease.

(25) The battery ammeter 34 is disposed between the bus line 21 and the converter 25 and measures a current flowing from the battery 26 toward the bus line 21. This current is hereinafter called a battery current. The battery ammeter 34 may be disposed between the converter 25 and the battery 26. In this case, a battery current IBm is corrected in accordance with the ratio of the voltage increase or the voltage decrease performed by the converter 25.

(26) The battery voltmeter 35 measures the voltage of the battery 26.

(27) The DC power system voltmeter 36 measures the voltage of the bus line 21. This measured voltage is hereinafter called a DC power system voltage.

(28) The converter controller 33 controls the operation of the converter 25 based on a DC power system voltage V_{Dm} and the battery current IB_m. The converter controller 33 is constituted by, for example, a calculator (not shown) including a processor (not shown) and a memory (not shown). One example of the calculator is a microcontroller. Examples of the processor include CPUs, MPUs, FPGAs (Field Programmable Gate Arrays), and PLCs (Programmable Logic Controllers). Examples of the memory include: internal memories of processors, such as ROMs and RAMs; and external memories, such as hard disc drives. The below-described control of the converter 25 is performed in such a manner that the processor reads and executes a predetermined converter control program stored in the memory.

(29) The power supply inverter 31 is one example of the bidirectional electric power converter. In Embodiment 1, the bidirectional electric power converter is used to mutually convert the AC power of the AC power system 10 and the DC power of the DC power system 20. Examples of the bidirectional electric power converter include bidirectional inverters and cycloconverters. In Embodiment 1, the bidirectional inverter is used as the power supply inverter 31. The power supply inverter 31 may be a voltage inverter or a current inverter. Herein, the voltage inverter is used as the power supply inverter 31.

(30) An AC terminal of the power supply inverter 31 is connected to the AC power system 10, and a DC terminal of the power supply inverter 31 is connected to the DC power system 20. The power

supply inverter **31** is controlled by the inverter controller **32** to mutually convert the AC power of the AC power system **10** and the DC power of the DC power system **20**. Moreover, the power supply inverter **31** is linked to (is operated in synchronization with) the AC power system **10**.
(31) The inverter controller **32** is one example of the bidirectional electric power conversion controller. The inverter controller **32** controls the operation of the power supply inverter **31** based on the battery current I_{Bm} and a battery voltage V_{Bm} .

(32) The inverter controller **32** is constituted by, for example, a calculator (not shown) including a processor (not shown) and a memory (not shown). One example of the calculator is a microcontroller. Examples of the processor include CPUs, MPUs, FPGAs (Field Programmable Gate Arrays), and PLCs (Programmable Logic Controllers). Examples of the memory include: internal memories of processors, such as ROMs and RAMs; and external memories, such as hard disc drives. The below-described control of the power supply inverter **31** is performed in such a manner that the processor reads and executes a predetermined converter control program stored in the memory. The inverter controller **32** and the converter controller **33** may be constituted by a single calculator.

Configuration of Inverter Controller **32**

Configuration Related to Electric Power Conversion Control

(33) First, the configuration related to electric power conversion control will be described. FIG. **2** is a block diagram showing the configuration related to the electric power conversion control of the inverter controller **32** of FIG. **1**. As shown in FIG. **2**, the inverter controller **32** includes a noise filter **41**, a subtracter **42**, a PI controller **43**, an inverter current controller **44**, a two-phase/three-phase converter **45**, and a three-phase/two-phase converter **46**.

(34) The noise filter **41** eliminates noise of the battery current I_{Bm} .

(35) The subtracter **42** subtracts the battery current I_{Bm} , from which the noise has been eliminated, from a predetermined target current value I_{Bt} of the battery current to generate a current deviation I_{Bte} . Herein, the predetermined target current value I_{Bt} of the battery current I_{Bm} is a current value that should be maintained when the battery current I_{Bm} is in a steady state.

(36) The PI controller **43** subjects the current deviation I_{Bte} to PI control to generate a d-axis current command value I_d . A d-axis represents an active electric power component of the AC power output from the power supply inverter **31**, and a q-axis represents a reactive electric power component of the AC power output from the power supply inverter **31**.

(37) The inverter current controller **44** generates a d-axis voltage command value V_d and a q-axis voltage command value V_q based on the d-axis current command value I_d , a q-axis current command value I_q , a d-axis current I_{df} , and a q-axis current I_{qf} . The d-axis current I_{df} and the q-axis current I_{qf} are input to the inverter current controller **44** from the three-phase/two-phase converter **46**. The q-axis current command value I_q is typically zero but may be a value that compensates reactive electric power generated by an LCL filter included in an AC output portion of the power supply inverter **31**. The d-axis current I_{df} and the q-axis current I_{qf} are currents obtained by converting measured values of a three-phase alternating current output by the power supply inverter **31** into two phases that are the d-axis and the q-axis. Specifically, the inverter current controller **44** subjects the current deviation of the d-axis current I_{df} with respect to the d-axis current command value I_d and the current deviation of the q-axis current I_{qf} with respect to the q-axis current command value I_q to predetermined control compensation to generate the d-axis voltage command value V_d and the q-axis voltage command value V_q .

(38) The two-phase/three-phase converter **68** converts the d-axis voltage command value V_d and the q-axis voltage command value V_q into three-phase voltage command values that are a U-phase voltage command value V_{uc} , a V-phase voltage command value V_{vc} , and a W-phase voltage command value V_{wc} and outputs the U-phase voltage command value V_{uc} , the V-phase voltage command value V_{vc} , and the W-phase voltage command value V_{wc} to the power supply inverter **31**.

(39) The power supply inverter **31** converts the DC voltage (DC power) of the DC power system **20** into the AC voltage (AC power) of the AC power system **10** based on the three-phase voltage command values.

(40) A three-phase current sensor (not shown) measures the current values of the converted AC power, and the measured three-phase current values (I_{uf} , I_{vf} , and I_{wf}) are converted into the d-axis current I_{df} and the q-axis current I_{qf} by the three-phase/two-phase converter **69**.

(41) As a result of this control, the DC power of the DC power system **20** is converted into the AC power of the AC power system **10** such that the battery current I_{Bm} coincides with the predetermined target current value I_{Bt} . When the d-axis current I_{df} is a positive value, the DC power of the DC power system **20** is converted into the AC power of the AC power system **10**. When the d-axis current I_{df} is a negative value, the AC power of the AC power system **10** is converted into the DC power of the DC power system **20**.

(42) As shown in FIG. 2, the subtracter **42** and the PI controller **43** constitute a controller (hereinafter referred to as a “battery current controller”) that controls the battery current I_{Bm} . The degree to which the battery **26** absorbs the load fluctuation changes depending on a response speed of the battery current controller (**42**, **43**). When the response speed of the battery current controller (**42**, **43**) is made high, the battery **26** absorbs only short-time, momentary load fluctuation. When the response speed of the battery current controller (**42**, **43**) is made low, the battery **26** absorbs the fluctuation having a time constant that is long to some extent. As above, a ratio at which the battery **26** absorbs the load fluctuation can be adjusted by adjusting the response speed of the battery current controller (**42**, **43**). The response speed of the battery current controller (**42**, **43**) can be adjusted by changing a control parameter of the PI controller **43**.

Configuration Related to Target Current Value Control of Battery Current

(43) Next, the configuration related to the control of the target current value I_{Bt} of the battery current I_{Bm} will be described. The inverter controller **32** adjusts the predetermined target current value I_{Bt} in accordance with a charge rate of the battery **26**. The charge rate of the battery **26** is represented by a ratio of a present voltage value to a voltage value in a fully charged state. The inverter controller **32** calculates, as a charge rate R_B , a ratio of a battery voltage value V_{Bm} to the voltage value in the fully charged state. Then, the predetermined target current value I_{Bt} is adjusted based on the charge rate R_B as below.

(44) In the present embodiment, a dead zone whose center corresponds to a target charge rate R_{Bt} is set for the charge rate R_B .

(45) When the charge rate R_B becomes the predetermined target charge rate R_{Bt} , the predetermined target current value I_{Bt} is set to zero. The target charge rate R_{Bt} is suitably determined in accordance with, for example, the specifications of the power supply system. Herein, the target charge rate R_{Bt} is set to, for example, 0.5. This setting can easily deal with both a sudden decrease and a sudden increase in the DC power load (**22**, **24**).

(46) When the charge rate R_B falls below a predetermined dead zone lower limit charge rate R_{Bll} , the predetermined target current value I_{Bt} is set to a negative value. The dead zone lower limit charge rate R_{Bll} is set to, for example, 0.4. In this case, the predetermined target current value I_{Bt} becomes a target current value of a charge current. With this, when the charge rate R_B of the battery **26** decreases by discharging, the charge rate R_B of the battery **26** can be made to return to the predetermined target charge rate R_{Bt} without the generation of hunting.

(47) When the charge rate R_B exceeds a predetermined dead zone upper limit charge rate R_{Bul} , the predetermined target current value I_{Bt} is set to a positive value. The dead zone upper limit charge rate R_{Bul} is set to, for example, 0.6. In this case, the predetermined target current value I_{Bt} becomes a target current value of a discharge current. With this, when the charge rate R_B of the battery **26** increases by charging, the charge rate R_B of the battery **26** can be made to return to the predetermined target charge rate R_{Bt} without the generating of hunting.

Configuration of Converter Controller **33**

- (48) FIG. 3 is a block diagram showing the configuration of the converter controller 33 of FIG. 1. As shown in FIG. 3, the converter controller 33 includes a noise filter 51, a subtracter 52, a PI controller 53, an inverting element 54, a subtracter 55, a PI controller 56, and a noise filter 57.
- (49) The noise filter 51 eliminates noise of the DC power system voltage V_{Dm} .
- (50) The subtracter 52 subtracts the DC power system voltage V_{Dm} , from which the noise has been eliminated, from a DC power system target voltage value V_{Dt} of the DC power system 20 to generate a voltage deviation V_{De} . The DC power system target voltage value V_{Dt} is the reference voltage value of the DC power system 20.
- (51) The PI controller 53 subjects the voltage deviation V_{De} to the PI control to generate a battery current command value I_{Bc} .
- (52) The noise filter 57 eliminates noise of the battery current I_{Bm} . The inverting element 54 inverts the positive or negative sign of the battery current I_{Bm} from which the noise has been eliminated. This is because in the control of the converter 25, the battery current during charging is positive.
- (53) The subtracter 55 subtracts the battery current I_{Bm} , whose positive or negative sign has been inverted, from the battery current command value I_{Bc} to generate a current deviation I_{Bce} .
- (54) The PI controller 5 subjects the current deviation I_{Bce} to the PI control to generate a duty command value D and outputs the duty command value D to the converter 25.
- (55) The converter 25 operates the DC chopper based on the duty command value D to increase or decrease the voltage of the battery 26. By this control, the battery 26 is discharged when the DC power system voltage V_{Dm} falls below the DC power system target voltage value V_{Dt} , and the battery 26 is charged when the DC power system voltage V_{Dm} exceeds the DC power system target voltage value V_{Dt} . With this, the DC power system voltage V_{Dm} is maintained at the DC power system target voltage value V_{Dt} .

Operation

- (56) Next, the operation of the electric power control system 100 configured as above will be described, i.e., the operation of the power supply system 200 will be described.
- (57) As shown in FIGS. 1 to 3, when sudden load fluctuation occurs at the DC power load (22 to 24), the DC power system voltage V_{Dm} fluctuates. In this case, the converter controller 33 controls the converter 25 to discharge or charge the battery 26 such that: the fluctuation of the DC power system voltage V_{Dm} is canceled; and the DC power system voltage V_{Dm} is maintained at the predetermined DC power system target voltage value V_{Dt} . With this, electric power corresponding to the load fluctuation is supplied to or discharged from the DC power load (22 to 24). As a result, the fluctuation of the voltage of the DC power system 20 is suppressed.
- (58) On the other hand, in this period of time, the inverter controller 32 controls the power supply inverter 31 to mutually convert the DC power of the DC power system 20 and the AC power of the AC power system 10 such that the battery current I_{Bm} coincides with the predetermined target current value I_{Bt} . Therefore, the discharging or charging of the battery 26 due to the sudden load fluctuation of the DC power load (22 to 24) converges after the discharging or charging by which the battery current I_{Bm} coincides with the predetermined target current value I_{Bt} that is stationary converges. As a result, the DC power of the DC power system 20 and the AC power of the AC power system 10 converge after the discharging or charging by which the battery current I_{Bm} coincides with the predetermined target current value I_{Bt} that is stationary converges. With this, the influence of the sudden load fluctuation of the DC power load (22 to 24) on the AC power system 10 is suppressed.
- (59) Moreover, when the charge rate R_B of the battery 26 changes due to the sudden load fluctuation of the DC power load (22 to 24), the inverter controller 32 adjusts the predetermined target current value I_{Bt} in accordance with the charge rate of the battery 26. As a result, the charge rate R_B of the battery 26 returns to the predetermined target charge rate R_{Bt} .

Simulation

(60) FIG. 4 is a graph showing the result of a simulation of the power supply system **200** of FIG. 1. FIG. 5 is a graph showing the result of a simulation of the power supply system of Comparative Example. Comparative Example is different from the power supply system **200** of Embodiment 1 in that in FIG. 1, the DC power system **20** does not include the electric power storage (**25**, **26**), and the power supply inverter **31** is replaced with a diode rectifier. Except for these, Comparative Example is the same in configuration as the power supply system **200** of Embodiment. In the power supply system **200** of Embodiment 1, the predetermined target current value I_{Bt} in the inverter controller **32** is set to zero.

(61) In this simulation, the DC power load (**22** to **24**) is increased and then decreased in a step manner in a time zone in which an elapsed time since the start of the simulation is between three seconds and seven seconds.

(62) In FIG. 4, horizontal axes of five graphs denote an elapsed time (seconds). A vertical axis of the uppermost graph denotes a DC voltage (DC power system voltage V_{Dm}) (V) of the DC power system **20**. A vertical axis of the second graph from the top denotes DC power (kW) in the DC power system **20**. A vertical axis of the third graph from the top denotes a frequency (Hz) of the AC power system **10**. A vertical axis of the fourth graph from the top denotes a voltage (V) of the AC power system **10**. A vertical axis of the lowermost graph denotes electric power (kW) of the AC power system **10**. Moreover, in the second graph from the top, a gray line labeled as “AC/DC” denotes DC power of the power supply inverter **31**, a one-dot chain line labeled as “DC/DC” denotes DC power of the converter **25**, and a broken line labeled as “DC Ld.” denotes DC power of the DC power load (**22** to **24**). Furthermore, in the lowermost graph, a gray line labeled as “Gen” denotes AC power of the power generator **12**, a one-dot chain line labeled as “AC/DC” denotes AC power of the power supply inverter **31**, and a broken line labeled as “AC Ld.” denotes AC power of the AC load (**14**).

(63) Five graphs in FIG. 5 are the same as the graphs in FIG. 4 except for the second graph from the top. A vertical axis of the second graph from the top denotes DC power (kW) in the DC power system **20** and the AC power system **10**. A gray line labeled as “AC/DC” denotes DC power of the diode rectifier, and a broken line labeled as “DC Ld.” denotes DC power of the DC power load (**22** to **24**).

(64) First, as shown in FIG. 5, in Comparative Example, due to the increase and decrease in the DC power load (**22** to **24**) in the step manner, the DC voltage suddenly decreases by about **50 V**. In addition, the fluctuation of the electric power is transmitted through the diode rectifier to the AC power system **10** as it is, and the load (AC power) of the power generator **12** driven by the engine suddenly fluctuates.

(65) On the other hand, as shown in FIG. 4, in the power supply system **200** of Embodiment 1, when the DC power load (**22** to **24**) suddenly fluctuates, the load fluctuation is absorbed by constant control of the DC voltage (DC power system voltage V_{Dm}) by the converter **25**. Then, the battery current I_{Bm} is controlled by the power supply inverter **31** to become zero, and therefore, the current (DC power) of the converter **25** converges to zero in about one to two seconds, and the load fluctuation (the AC power of the power supply inverter **31**) transmitted to the AC power system **10** gently rises. With this, the DC voltage of the DC power system **20** is significantly suppressed, and the load fluctuation of the power generator **12** driven by the engine and the fluctuation of the AC voltage and frequency of the AC power system **10** become gentle.

(66) In this simulation, in the control of the battery current I_{Bm} by the power supply inverter **31**, the control parameter is set such that the battery current I_{Bm} converges to zero with a time constant of about 0.6 second. However, the time constant can be adjusted by changing the control parameter. Therefore, the adjustment (selection) of the time constant together with the selection of the necessary capacity of the battery **26** can contribute to the optimal design of the power supply system **200**.

Embodiment 2

(67) As an example, Embodiment 2 of the present application describes a case where second control of suppressing the influence of the load fluctuation on the AC power system **10** is superimposed on the first control of the power supply system **200** of Embodiment 1.

(68) As shown in FIGS. **1** and **2**, the first control in the power supply system **200** of Embodiment 1 is to control the output or input of the AC terminal of the bidirectional electric power converter (**31**) such that the charge current, discharge current, charge electric power, or discharge electric power of the electric power storage (**25**, **26**) coincides with a predetermined target current value I_{Bt} (see FIG. **2**) or a predetermined target electric power value. In Embodiment 2, the bidirectional electric power conversion controller (**31**) performs the second control of controlling the output or input of the AC terminal of the bidirectional electric power converter (**31**) so as to suppress the fluctuation of at least one of the electric power, voltage, current, or frequency of the AC power system **10** such that the second control is superimposed on the first control. Herein, the above expression “the second control is superimposed on the first control” denotes that “the output or input of the AC terminal of the bidirectional electric power converter is controlled such that: the charge current, discharge current, charge electric power, or discharge electric power of the electric power storage (**25**, **26**) coincides with the predetermined target current value I_{Bt} or the predetermined target electric power value; and the fluctuation of at least one of the electric power, voltage, current, or frequency of the AC power system **10** is suppressed.” Moreover, the cause of “the fluctuation of at least one of the electric power, voltage, current, or frequency of the AC power system **10**” is not especially limited, and examples of the cause include the load fluctuation of the AC power load (**14**) of the AC power system **10** and the load fluctuation of the DC power load (**22** to **24**) of the DC power system **20**.

(69) An object of the second control is to, when at least one of the electric power, voltage, current, or frequency of the AC power system **10** fluctuates due to the load fluctuation, suppress the fluctuation of at least one of the electric power, voltage, current, or frequency of the AC power system **10** in such a manner that the electric power (or the current) is supplied to or discharged from the AC power system **10** by the bidirectional electric power converter (**31**).

(70) As an example, the following will describe a case where the second control controls the output or input of the AC side of the bidirectional electric power converter (**31**) so as to suppress the fluctuation of the frequency of the AC power system **10** by feedback of active electric power output from or input to the AC side of the bidirectional electric power converter (**31**). Moreover, as an example, each of Embodiments 5 and 6 describes a case where the second control controls the output or input of the AC side of the bidirectional electric power converter (**31**) so as to suppress the fluctuation of the voltage of the AC power system **10** by feedback of the reactive electric power or active electric power output from or input to the AC side of the bidirectional electric power converter (**31**). Cases of the second control other than the above cases can be performed in the same manner as Embodiments 2, 5, and 6 with reference to Embodiments 2, 5, and 6.

(71) Moreover, when performing intended control instead of stabilizing the AC power system **10**, in addition to the suppression of the fluctuation, the second control of controlling the output or input of the AC terminal of the bidirectional electric power converter (**31**) may be performed based on at least one of the electric power, voltage, current, or frequency of the AC power system **10** so as to be superimposed on the first control.

(72) Hereinafter, Embodiment 2 will be specifically described.

Specific Configuration and Operation

(73) FIG. **6** is a block diagram showing the configuration of the power supply system according to Embodiment 2 of the present application.

(74) As shown in FIG. **6**, the power supply system **200** of Embodiment 2 further includes an AC power system voltmeter **71**. The AC power system voltmeter **71** measures the voltage of the bus line **13** of the AC power system **10**. Hereinafter, this measured voltage is called an AC power system voltage.

(75) FIG. 7 is a block diagram showing the configuration of the inverter controller 32 of FIG. 6. As shown in FIG. 7, the inverter controller 32 includes the noise filter 41, the subtracter 42, a PI controller 61, a noise filter 62, a drooping unit 63, an adder-subtractor 64, a subtracter 65, a PI controller 66, the inverter current controller 44, the two-phase/three-phase converter 45, and the three-phase/two-phase converter 46.

(76) The noise filter 41 and the subtracter 42 are the same as the noise filter 41 and the subtracter 42 in Embodiment 1, respectively. Moreover, the inverter current controller 44, the two-phase/three-phase converter 45, and the three-phase/two-phase converter 46 are the same as the inverter current controller 44, the two-phase/three-phase converter 45, and the three-phase/two-phase converter 46 in Embodiment 1, respectively.

(77) The PI controller 61 subjects the current deviation IB_{te} , generated by the subtracter 42, to the PI control to generate a target frequency value F_i corresponding to the battery current IB_m .

(78) On the other hand, the noise filter 62 eliminates noise of the d-axis current I_{df} converted by the three-phase/two-phase converter 46. The drooping unit 63 converts the d-axis current I_{df} , from which the noise has been eliminated, into a target frequency value F_d corresponding to the d-axis current in accordance with a drooping characteristic.

(79) The adder-subtractor 64 generates a total target frequency value F_s by: adding the target frequency value F_i corresponding to the battery current IB_m to a predetermined target frequency value F_t that is a reference frequency of the AC power system 10; and subtracting the target frequency value F_d corresponding to the d-axis current from the predetermined target frequency value F_t .

(80) On the other hand, the inverter controller 32 generates an AC power frequency F_m based on the AC power system voltage measured by the AC power system voltmeter 71.

(81) The subtracter 65 generates a frequency deviation F_e by subtracting an AC power system frequency F_m from the total target frequency value F_s .

(82) The PI controller 66 generates the d-axis current command value I_d by subjecting the frequency deviation F_e to the PI control. Since subsequent processing is the same as that of the inverter controller 32 of Embodiment 1, the explanation thereof is omitted.

(83) According to this control, in a process in which the DC power of the DC power system 20 and the AC power of the AC power system 10 are mutually converted such that the battery current IB_m coincides with the predetermined target current value IB_t , the d-axis current that suppresses the fluctuation of the frequency F_m of the AC power system 10 in accordance with the fluctuation of the frequency F_m is output from or input to the AC side of the power supply inverter 31. This is because when the frequency F_m of the AC power system 10 fluctuates to deviate from the target frequency value F_t , the d-axis current that cancels this fluctuation is output from or input to the AC side of the power supply inverter 31. The d-axis current represents the active electric power.

(84) As a result, the fluctuation of the frequency F_m of the AC power system 10 is suppressed. In this control, the frequency F_m of the AC power system 10 does not converge to the target frequency value F_t . This is because the target frequency value F_d corresponding to the d-axis current I_d changes by the output of the power supply inverter 31 itself, and therefore, the total target frequency value F_s changes. However, this is not a problem. This is because an object of this control is simply to, when the frequency F_m fluctuates due to the load fluctuation, suppress the fluctuation of the frequency F_m in such a manner that the electric power (or the current) is supplied to or discharged from the AC power system 10 by the power supply inverter 31.

(85) When the d-axis current I_{df} is a positive value, the DC power of the DC power system 20 is converted into the AC power of the AC power system 10. When the d-axis current I_{df} is a negative value, the AC power of the AC power system 10 is converted into the DC power of the DC power system 20.

(86) Herein, the following points are important.

(87) In Embodiment 2, the response of the second control is set faster than the response of the first

control. The response speed of the first control is set by the control parameter of the PI controller **61**, and the response speed of the second control is set by the control parameter of the PI controller **66**.

(88) For example, when the load of the AC load **14** fluctuates, and the power supply inverter **31** absorbs this fluctuation, this load fluctuation is absorbed by the electric power storage (**25**, **26**) through the power supply inverter **31**. However, if the electric power storage (**25**, **26**) tries to absorb this load fluctuation after a charge or discharge current I_{Bm} of the electric power storage (**25**, **26**) converges by the first control so as to coincide with the predetermined target current value I_{Bt} , the charge or discharge current I_{Bm} is maintained at the predetermined target current value I_{Bt} , and therefore, the first control cancels the second control, and the electric power storage (**25**, **26**) cannot absorb this load fluctuation. However, according to this configuration, since the response of the second control is faster than the response of the first control, the first control does not cancel the second control, and the electric power storage (**25**, **26**) can absorb the load fluctuation of the AC load **14**. However, that the response of the second control is faster than the response of the first control is not essential in Embodiment 2. This is because when the second control is superimposed on the first control, the load fluctuation is absorbed by the electric power storage (**25**, **26**) through the power supply inverter **31** by the second control until the charge or discharge current I_{Bm} of the electric power storage (**25**, **26**) converges by the first control so as to coincide with the predetermined target current value I_{Bt} .

(89) As described above, according to Embodiment 2, the load fluctuation is shared with the power generator **12**, and the load fluctuation of the power generator **12** driven by the engine and the fluctuation of the frequency F_m of the AC power system **10** can be suppressed.

Simulation

(90) FIG. **8** is a graph showing the result of a simulation of the power supply system **200** of FIG. **6**. Comparative Example of Embodiment 2 is the same as Comparative Example of Embodiment **1**. How to draw the graphs in FIG. **8** is the same as how to draw the graphs in FIG. **4**. How to draw the graphs in FIG. **9** is the same as how to draw the graphs in FIG. **5**.

(91) In the power supply system **200** of Embodiment 2, the predetermined target current value I_{Bt} in the inverter controller **32** is set to zero. Moreover, a control gain is set such that a convergence time constant of the battery current I_{Bm} of the converter **25** becomes about **1.5** seconds.

(92) In this simulation, the DC power load (**22** to **24**) is applied after three seconds from the start of the simulation, and the DC power load (**22** to **24**) is cut after seven seconds from the start of the simulation. Then, the AC load (**14**) is applied after 12 seconds from the start of the simulation, and the AC load (**14**) is cut after **16** seconds from the start of the simulation.

(93) First, as shown in FIG. **9**, in Comparative Example, the fluctuation when the DC power load (**22** to **24**) is applied is similar to that of FIG. **5**. As with when the DC power load (**22** to **24**) is applied, when the AC load (**14**) is applied, the frequency and voltage of the AC power system fluctuate, and the load (AC power) of the power generator **12** driven by the engine suddenly fluctuates.

(94) On the other hand, as shown in FIG. **8**, the behavior when the DC power load (**22** to **24**) is applied is substantially similar to that of FIG. **4** although a convergence time is different since the control gain is set such that the convergence time constant of the battery current I_{Bm} of the converter **25** becomes about **1.5** seconds. On the other hand, when the AC load (**14**) is applied, the steep load fluctuation is absorbed by the power supply inverter **31** having been subjected to the drooping control, and with this, the fluctuation of the load (AC power) of the power generator **12** is suppressed, and the fluctuation of the frequency and voltage of the AC power system **10** is significantly suppressed. The entire load fluctuation absorbed by the power supply inverter **31** by the drooping control is absorbed by the converter **25** through the DC power system **20**. Therefore, after the load fluctuation is absorbed, the battery current I_{Bm} of the converter **25** converges to zero with a time constant of about 1.5 seconds again, and the power generator **12** absorbs all the

stationary loads.

Embodiment 3

(95) According to Embodiment 3 of the present application, a dead zone is set for the battery current I_{Bm} of the battery current controller of Embodiment 2.

(96) As shown in FIG. 7, the subtracter **42** and the PI controller **61** constitute the battery current controller. FIG. 10 is a flow chart showing a dead zone operation of the battery current controller (**42**, **61**) of the inverter controller of the power supply system according to Embodiment 3 of the present application.

(97) As shown in FIG. 10, in a case where the battery current I_{Bm} exceeds the predetermined target current value I_{Bt} of the battery current I_{Bm} by a fixed value I_{Boft} (Yes in Step **S1**), the battery current controller (**42**, **61**) decreases the target frequency value F_i corresponding to the battery current I_{Bm} at a fixed rate (ΔF_i [Hz/sec]) (Step **S2**). In a case where the battery current I_{Bm} falls below the predetermined target current value I_{Bt} of the battery current I_{Bm} by the fixed value I_{Boft} (Yes in Step **S3**), the battery current controller (**42**, **61**) increases the target frequency value F_i corresponding to the battery current I_{Bm} at the fixed rate (ΔF_i [Hz/sec]) (Step **S4**). In a case other than the above cases (No in Step **S3**), the battery current controller (**42**, **61**) maintains the target frequency value F_i corresponding to the battery current I_{Bm} at a previous value (Step **S5**).

(98) With this, the operation of generating the target frequency value F_i corresponding to the battery current I_{Bm} is stabilized (hunting is prevented).

(99) FIG. 11 is a graph showing the result of a simulation of the power supply system **200** of Embodiment 3. How to draw the graphs of FIG. 11 is the same as that of FIG. 8. Conditions of the simulation of Embodiment 3 are the same as the conditions of the simulation of Embodiment 2.

(100) As shown in FIG. 11, results similar to those of the simulation of Embodiment 2 are obtained.

Embodiment 4

(101) As an example, Embodiment 4 of the present application describes a case where in the electric power control system **100** (power supply system **200**) of Embodiment 2, the second control further includes control of controlling the output or input of the AC side of the bidirectional electric power converter (**31**) so as to suppress the fluctuation of the voltage of the AC power system **10** by feedback of the reactive electric power output from or input to the AC side of the bidirectional electric power converter (**31**).

(102) FIG. 12 is a block diagram showing the configuration of the inverter controller **32** of the power supply system **200** according to Embodiment 4 of the present application.

(103) As shown in FIG. 12, the inverter controller **32** of Embodiment 4 corresponds to the inverter controller **32** of Embodiment 2 and further includes a noise filter **81**, a drooping unit **82**, a subtracter **83**, a subtracter **84**, and a PI controller **85**.

(104) The noise filter **81** eliminates the noise of the q-axis current I_{qf} converted by the three-phase/two-phase converter **46**. The drooping unit **82** converts the q-axis current I_{qf} , from which the noise has been eliminated, into a target AC voltage value V_{Ai} corresponding to the q-axis current in accordance with the drooping characteristic.

(105) The subtracter **83** generates a total target AC voltage value V_A s by subtracting the target AC voltage value V_{Ai} corresponding to the q-axis current from a predetermined target AC voltage value V_{At} that is a reference voltage of the AC power system **10**.

(106) The subtracter **84** generates a voltage deviation V_{Ae} by subtracting an AC power system voltage V_{Am} , measured by the AC power system voltmeter **71**, from the total target AC voltage value V_A s.

(107) The PI controller **85** generates the q-axis current command value I_q by subjecting the voltage deviation V_{Ae} to the PI control. Since subsequent processing is the same as that of the inverter controller **32** of Embodiment 1, the explanation thereof is omitted.

(108) As a result of the second control, in addition to the effects of Embodiment 2, in a process in which the DC power of the DC power system **20** and the AC power of the AC power system **10** are

mutually converted such that the battery current I_{Bm} coincides with the predetermined target current value I_{Bt} , the q-axis current that suppresses the fluctuation of the voltage V_{Am} of the AC power system **10** in accordance with the fluctuation of the voltage V_{Am} is output from or input to the AC side of the power supply inverter **31**. This is because when the voltage V_{Am} of the AC power system **10** fluctuates to deviate from the target AC voltage value V_{At} , the q-axis current that cancels this fluctuation is output from or input to the AC side of the power supply inverter **31**. The q-axis current represents the reactive electric power.

(109) As a result, the fluctuation of the voltage V_{Am} of the AC power system **10** is suppressed.

(110) As described above, according to Embodiment **4**, the load fluctuation is shared with the power generator **12**, and the load fluctuation of the power generator **12** driven by the engine and the fluctuation of the frequency F_m and voltage of the AC power system **10** can be suppressed.

Embodiment 5

(111) As an example, Embodiment 5 of the present application describes a case where in the electric power control system **100** (power supply system **200**) of Embodiment 1, the second control controls the output or input of the AC side of the bidirectional electric power converter (**31**) so as to suppress the fluctuation of the voltage of the AC power system **10** by feedback of the reactive electric power output from or input to the AC side of the bidirectional electric power converter (**31**).

(112) FIG. **13** is a block diagram showing the configuration of the inverter controller **32** of the power supply system according to Embodiment 5 of the present application.

(113) As shown in FIG. **13**, the inverter controller **32** of Embodiment 5 corresponds to the inverter controller **32** of Embodiment 1 and further includes the noise filter **81**, the drooping unit **82**, the subtracter **83**, the subtracter **84**, and the PI controller **85**.

(114) Since the configurations of these additional components are the same as the configurations of the additional components added to Embodiment 2 in Embodiment 4, the explanations thereof are omitted.

(115) According to the second control in Embodiment 5, in a process in which the DC power of the DC power system **20** and the AC power of the AC power system **10** are mutually converted such that the battery current I_{Bm} coincides with the predetermined target current value I_{Bt} , the q-axis current that suppresses the fluctuation of the voltage V_{Am} of the AC power system **10** in accordance with the fluctuation of the voltage V_{Am} is output from or input to the AC side of the power supply inverter **31**. As a result, the fluctuation of the voltage V_{Am} of the AC power system **10** is suppressed.

(116) Therefore, according to Embodiment 5, in addition to the effects of Embodiment 1, the fluctuation of the voltage V_{Am} of the AC power system **10** can be suppressed.

Embodiment 6

(117) As an example, Embodiment 6 of the present application describes a case where in the electric power control system **100** (power supply system **200**) of Embodiment 1, the second control controls the output or input of the AC side of the bidirectional electric power converter (**31**) so as to suppress the fluctuation of the voltage of the AC power system **10** by feedback of the active electric power output from or input to the AC side of the bidirectional electric power converter (**31**).

(118) FIG. **14** is a block diagram showing the configuration of the inverter controller **32** of the power supply system according to Embodiment 6 of the present application.

(119) As shown in FIG. **14**, the inverter controller **32** of Embodiment 6 corresponds to the inverter controller **32** of Embodiment 1 and further includes the noise filter **81**, the drooping unit **82**, the subtracter **83**, the subtracter **84**, the PI controller **85**, and an adder **86**. The configurations and operations of the noise filter **41**, the subtracter **42**, and the PI controller **43** are the same as those of Embodiment 1. Herein, the d-axis current command value output from the PI controller **43** is represented by a first d-axis current command value I_{dl} .

(120) The noise filter **81** eliminates the noise of the d-axis current I_{df} converted by the three-

phase/two-phase converter **46**. The drooping unit **82** converts the d-axis current I_{df} , from which the noise has been eliminated, into the target AC voltage value V_{Ai} corresponding to the d-axis current in accordance with the drooping characteristic.

(121) The subtracter **83** generates the total target AC voltage value V_A s by subtracting the target AC voltage value V_{Ai} corresponding to the d-axis current from the predetermined target AC voltage value V_{At} that is the reference voltage of the AC power system **10**.

(122) The subtracter **84** generates the voltage deviation V_{Ae} by subtracting the AC power system voltage V_{Am} , measured by the AC power system voltmeter **71**, from the total target AC voltage value V_A s.

(123) The PI controller **85** generates a second d-axis current command value I_{d2} by subjecting the voltage deviation V_{Ae} to the PI control.

(124) The adder **86** generates the d-axis current command value I_d by adding the second d-axis current command value I_{d2} to the first d-axis current command value I_{d1} .

(125) Since subsequent processing is the same as that of the inverter controller **32** of Embodiment 1, the explanation thereof is omitted.

(126) According to the second control of Embodiment 6 configured as above, in a process in which the DC power of the DC power system **20** and the AC power of the AC power system **10** are mutually converted such that the battery current I_{Bm} coincides with the predetermined target current value I_{Bt} , the d-axis current that suppresses the fluctuation of the voltage V_{Am} of the AC power system **10** in accordance with the fluctuation of the voltage V_{Am} is output from or input to the AC side of the power supply inverter **31**. This is because when the voltage V_{Am} of the AC power system **10** fluctuates to deviate from the target AC voltage value V_{At} , the d-axis current that cancels this fluctuation is output from or input to the AC side of the power supply inverter **31**. The d-axis current represents the active electric power.

(127) As a result, the fluctuation of the voltage V_{Am} of the AC power system **10** is suppressed.

(128) As described above, according to Embodiment 6, the load fluctuation is shared with the power generator **12**, and the load fluctuation of the power generator **12** driven by the engine and the fluctuation of the voltage V_{Am} of the AC power system **10** can be suppressed.

Other Embodiments

(129) In Embodiments 1 to 5, the converter **25** and the converter controller **33** may be omitted. Even in this case, the fluctuation of the voltage of the DC power system **20** due to the load fluctuation can be suppressed.

(130) In Embodiments 1 to 6, for example, the second control of extracting only a fluctuation component from the measured value of the frequency of the AC power system **10** through a high-pass filter and controlling the fluctuation component to zero may be added.

(131) From the foregoing description, numerous modifications and other embodiments of the present disclosure are obvious to those skilled in the art. Accordingly, the foregoing description is to be construed as illustrative only.

(132) The functionality of the elements disclosed herein may be implemented using circuitry or processing circuitry which includes general purpose processors, special purpose processors, integrated circuits, ASICs ("Application Specific Integrated Circuits"), conventional circuitry and/or combinations thereof which are configured or programmed to perform the disclosed functionality. Processors are considered processing circuitry or circuitry as they include transistors and other circuitry therein. The processor may be a programmed processor which executes a program stored in a memory. In the disclosure, the circuitry, units, or means are hardware that carry out or are programmed to perform the recited functionality. The hardware may be any hardware disclosed herein or otherwise known which is programmed or configured to carry out the recited functionality. When the hardware is a processor which may be considered a type of circuitry, the circuitry, means, or units are a combination of hardware and software, the software being used to configure the hardware and/or processor.

(133) An electric power control system according to one aspect of the present application includes: a bidirectional electric power converter including an AC terminal and a DC terminal, the AC terminal being connected to an AC power system, the DC terminal being connected to a DC power system to which one or more DC power loads are connected, the bidirectional electric power converter mutually converting AC power of the AC power system and DC power of the DC power system; an electric power storage that is connected to the DC power system and stores electric power of the DC power system; and a bidirectional electric power conversion controller that performs first control of controlling an output or input of the AC terminal of the bidirectional electric power converter such that a charge current, discharge current, charge electric power, or discharge electric power of the electric power storage coincides with a predetermined target current value or a predetermined target electric power value. Herein, “controlling an output or input of the AC terminal of the bidirectional electric power converter” denotes controlling voltage output, current output, electric power output, voltage input, current input, or electric power input of the AC terminal of the bidirectional electric power converter.

(134) According to this configuration, when sudden load fluctuation occurs at the DC power load, the electric power storage is charged or discharged such that the fluctuation of the voltage of the DC power system due to the sudden load fluctuation is canceled. With this, the fluctuation of the voltage of the DC power system is suppressed. On the other hand, in this period of time, the bidirectional electric power conversion controller controls the output or input of the AC terminal of the bidirectional electric power converter such that the charge current, discharge current, charge electric power, or discharge electric power of the electric power storage coincides with a predetermined target current value or a predetermined target electric power value. Therefore, the charging or discharging of the electric power storage due to the sudden load fluctuation of the DC power load converges after the charging or discharging by which the charge current, discharge current, charge electric power, or discharge electric power of the electric power storage coincides with a stationary predetermined target current value or a stationary predetermined target electric power value converges. As a result, the DC power of the DC power system and the AC power of the AC power system converge in a stationary state after the charging or discharging by which the charge current, discharge current, charge electric power, or discharge electric power of the electric power storage coincides with a stationary predetermined target current value or a stationary predetermined target electric power value converges. With this, the influence of the sudden load fluctuation of the DC power load on the AC power system is suppressed. Moreover, a ratio at which the electric power storage absorbs the load fluctuation (a ratio between the load fluctuation absorbed by the electric power storage and the load fluctuation absorbed by an electric power source (for example, the power generator) of the AC power system) can be adjusted by adjusting the response speed of the bidirectional electric power conversion controller.

(135) The predetermined target current value or the predetermined target electric power value may be zero.

(136) According to this configuration, the charge rate of the electric power storage is maintained constant.

(137) When a charge rate of the electric power storage becomes a predetermined target charge rate, the bidirectional electric power conversion controller may set the predetermined target current value or the predetermined target electric power value to zero.

(138) According to this configuration, the charge rate of the electric power storage can be maintained at the predetermined target charge rate.

(139) When the charge rate of the electric power storage falls below a predetermined dead zone lower limit charge rate that is lower than the target charge rate, the bidirectional electric power conversion controller may set the predetermined target current value or the predetermined target electric power value to a target current value of the charge current or a target electric power value of the charge electric power.

(140) According to this configuration, when the charge rate of the electric power storage decreases by discharging, the charge rate of the electric power storage can be made to return to the predetermined target charge rate without the generation of hunting.

(141) When the charge rate of the electric power storage exceeds a predetermined dead zone upper limit charge rate that is higher than the target charge rate, the bidirectional electric power conversion controller may set the predetermined target current value or the predetermined target electric power value to a target current value of the discharge current or a target electric power value of the discharge electric power.

(142) According to this configuration, when the charge rate of the electric power storage increases by charging, the charge rate of the electric power storage can be made to return to the predetermined target charge rate without the generation of hunting.

(143) The electric power storage may include: an electric power storage device that stores the electric power of the DC power system; and a DC/DC converter that controls a ratio of an increase or decrease of a voltage of the electric power storage device such that a voltage of the DC power system is maintained at a predetermined target voltage value.

(144) According to this configuration, when the fluctuation of the voltage of the DC power system occurs due to the sudden load fluctuation of the DC power load, the DC/DC converter controls the ratio of the increase or decrease of the voltage of the electric power storage device such that the voltage of the DC power system is maintained at the predetermined target voltage value. Therefore, the voltage of the DC power system quickly returns to the predetermined target voltage value.

(145) The electric power storage may include an electric power storage device that is directly connected to the DC power system and stores the electric power of the DC power system.

(146) According to this configuration, when the sudden load fluctuation occurs at the DC power load, the electric power storage device is charged or discharged such that the fluctuation of the voltage of the DC power system due to the sudden load fluctuation is canceled. With this, the fluctuation of the voltage of the DC power system is suppressed.

(147) The bidirectional electric power conversion controller may perform second control of controlling the output or input of the AC terminal of the bidirectional electric power converter so as to suppress a fluctuation of at least one of electric power, voltage, current, or frequency of the AC power system such that the second control is superimposed on the first control. Herein, the above expression “the second control is superimposed on the first control” denotes that “the output or input of the AC terminal of the bidirectional electric power converter is controlled such that: the charge current, discharge current, charge electric power, or discharge electric power of the electric power storage coincides with the predetermined target current value or the predetermined target electric power value; and the fluctuation of at least one of the electric power, voltage, current, or frequency of the AC power system is suppressed.” Moreover, the cause of “the fluctuation of at least one of the electric power, voltage, current, or frequency of the AC power system” is not especially limited, and examples of the cause include the AC power load of the AC power system and the DC power load of the DC power system **20**.

(148) According to this configuration, the fluctuation of at least one of the electric power, voltage, current, or frequency of the AC power system can be suppressed by the second control.

(149) A response of the second control may be faster than a response of the first control.

(150) When the AC load fluctuates, and the bidirectional electric power converter absorbs this fluctuation, this load fluctuation is absorbed by the electric power storage through the bidirectional electric power converter. However, if the electric power storage tries to absorb this load fluctuation after the charge current, discharge current, charge electric power, or discharge electric power of the electric power storage converges by the first control so as to coincide with the predetermined target current value or the predetermined target electric power value, the charge current, discharge current, charge electric power or discharge electric power is maintained at the predetermined target current value or the predetermined target electric power value, and therefore, the first control

cancels the second control, and the electric power storage cannot absorb the load fluctuation. However, according to this configuration, since the response of the second control is faster than the response of the first control, the first control does not cancel the second control, and the electric power storage can absorb the fluctuation of the AC load.

(151) The second control may include controlling the output or input of the AC side of the bidirectional electric power converter so as to suppress the fluctuation of the frequency of the AC power system by feedback of active electric power output from or input to the AC side of the bidirectional electric power converter.

(152) According to this configuration, the active electric power that suppresses the fluctuation of the frequency of the AC power system in accordance with the fluctuation of the frequency is output from or input to the AC side of the bidirectional electric power converter, and with this, the fluctuation of the frequency is suppressed.

(153) The second control may include controlling the output or input of the AC side of the bidirectional electric power converter so as to suppress the fluctuation of the voltage of the AC power system by feedback of reactive electric power or active electric power output from or input to the AC side of the bidirectional electric power converter.

(154) According to this configuration, the active electric power or reactive electric power that suppresses the fluctuation of the voltage of the AC power system in accordance with the fluctuation of the voltage is output from or input to the AC side of the bidirectional electric power converter, and with this, the fluctuation of the voltage is suppressed.

(155) Moreover, a power supply system according to another aspect of the present application includes any one of the above electric power control systems, the AC power system, and the DC power system.

(156) According to this configuration, the influence of the sudden load fluctuation of the DC power load on the AC power system can be suppressed.

INDUSTRIAL APPLICABILITY

(157) The electric power control system and the power supply system of the present application are useful as an electric power control system and a power supply system, each of which is able to, when sudden load fluctuation occurs at a general DC power load, suppress an influence of the load fluctuation on an AC power system.

Claims

1. An electric power control system comprising: a bidirectional electric power converter including an AC terminal and a DC terminal, the AC terminal being connected to an AC power system, the DC terminal being connected to a DC power system to which one or more DC power loads are connected, the bidirectional electric power converter mutually converting AC power of the AC power system and DC power of the DC power system; an electric power storage that is connected to the DC power system and stores electric power of the DC power system; and a bidirectional electric power conversion controller that performs first control of controlling an output or input of the AC terminal of the bidirectional electric power converter such that a charge current, discharge current, charge electric power, or discharge electric power of the electric power storage coincides with a predetermined target current value or a predetermined target electric power value, wherein the bidirectional electric power conversion controller performs second control of controlling the output or input of the AC terminal of the bidirectional electric power converter to suppress a fluctuation of at least one of electric power, voltage, current, or frequency of the AC power system such that the second control is superimposed on the first control.

2. The electric power control system according to claim 1, wherein the predetermined target current value or the predetermined target electric power value is zero.

3. The electric power control system according to claim 1, wherein when a charge rate of the

- electric power storage becomes a predetermined target charge rate, the bidirectional electric power conversion controller sets the predetermined target current value or the predetermined target electric power value to zero.
4. The electric power control system according to claim 3, wherein when the charge rate of the electric power storage falls below a predetermined dead zone lower limit charge rate that is lower than the target charge rate, the bidirectional electric power conversion controller sets the predetermined target current value or the predetermined target electric power value to a target current value of the charge current or a target electric power value of the charge electric power.
5. The electric power control system according to claim 3, wherein when the charge rate of the electric power storage exceeds a predetermined dead zone upper limit charge rate that is higher than the target charge rate, the bidirectional electric power conversion controller sets the predetermined target current value or the predetermined target electric power value to a target current value of the discharge current or a target electric power value of the discharge electric power.
6. The electric power control system according to claim 1, wherein the electric power storage includes: an electric power storage device that stores the electric power of the DC power system; and a DC/DC converter that controls a ratio of an increase or decrease of a voltage of the electric power storage device such that a voltage of the DC power system is maintained at a predetermined target voltage value.
7. The electric power control system according to claim 1, wherein the electric power storage includes an electric power storage device that is directly connected to the DC power system and stores the electric power of the DC power system.
8. The electric power control system according to claim 1, wherein a response of the second control is faster than a response of the first control.
9. The electric power control system according to claim 1, wherein the second control includes controlling the output or input of the AC side of the bidirectional electric power converter so as to suppress the fluctuation of the frequency of the AC power system by feedback of active electric power output from or input to the AC side of the bidirectional electric power converter.
10. The electric power control system according to claim 1, wherein the second control includes controlling the output or input of the AC side of the bidirectional electric power converter so as to suppress the fluctuation of the voltage of the AC power system by feedback of reactive electric power or active electric power output from or input to the AC side of the bidirectional electric power converter.
11. A power supply system comprising: the electric power control system according to claim 1; the AC power system; and the DC power system.
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